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Q3

Input = "Not a good plist"

T = 4

$$x_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad x_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix},$$

$$x_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

weights and Biases

$$W_{hh} = \begin{bmatrix} 0.5 & -0.2 \\ 0.1 & 0.6 \end{bmatrix}$$

$$W_{xh} = \begin{bmatrix} -0.9 & 0.1 & 0.8 & 0.2 \\ 0.7 & -0.1 & -0.6 & 0.3 \end{bmatrix}$$

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$W_{hy} = [1.5 \quad -1.2]$; $b_h = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$b_y = 0.5$; $h_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Standard RNN

$$a_t = W_{hh} h_{t-1} + W_{xh} X_t + b_h$$

$$h_t = \tanh(a_t)$$

Multiplying $W_{xh} X_t$

- $W_{xh} X_1 = \begin{bmatrix} -0.9 \\ 0.7 \end{bmatrix}$
- $W_{xh} X_2 = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}$
- $W_{xh} X_3 = \begin{bmatrix} 0.8 \\ -0.6 \end{bmatrix}$
- $W_{xh} X_4 = \begin{bmatrix} 0.2 \\ 0.3 \end{bmatrix}$

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S1 Word x_1 = "Not"

$$a_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -0.9 \\ 0.7 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -0.9 \\ 0.7 \end{bmatrix}$$

$$h_1 = \tanh \begin{bmatrix} -0.9 \\ 0.7 \end{bmatrix}$$

$$= \begin{bmatrix} -0.7163 \\ 0.6044 \end{bmatrix}$$

S2 Word x_2 = "a"

$$W_{hh} h_1 = \begin{bmatrix} 0.5(-0.7163) - 0.2(0.6044) \\ 0.1(-0.7163) - 0.6(0.6044) \end{bmatrix}$$

$$= \begin{bmatrix} -0.4791 \\ 0.0290 \end{bmatrix}$$

$$h_2 = \tanh \begin{bmatrix} -0.3791 \\ 0.1910 \end{bmatrix} = \begin{bmatrix} -0.3620 \\ 0.1887 \end{bmatrix}$$

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S3 (word X_3 = "good"

$$W_{hh} h_2 = \begin{bmatrix} 0.5(-0.3620) - 0.2(0.1887) \\ 0.1[-0.3620) + 0.6(0.1887] \end{bmatrix}$$

$$= \begin{bmatrix} -0.2187 \\ 0.0770 \end{bmatrix}$$

$$a_3 = \begin{bmatrix} -0.2187 \\ 0.0770 \end{bmatrix} + \begin{bmatrix} 0.8 \\ -0.6 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5813 \\ -0.5230 \end{bmatrix}$$

$$h_3 = \tanh \begin{bmatrix} 0.5813 \\ -0.5230 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5236 \\ -0.4799 \end{bmatrix}$$

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step 4 word x_4 = "play"

$$w_{hh} h_3 = \begin{bmatrix} 0.5(0.5236) - 0.2(-0.4799) \\ 0.1(0.5236) + 0.6(-0.4799) \end{bmatrix}$$

$$= \begin{bmatrix} 0.3578 \\ -0.2355 \end{bmatrix}$$

$$a_4 = \begin{bmatrix} 0.3578 \\ -0.2355 \end{bmatrix} + \begin{bmatrix} 0.2 \\ 0.3 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5578 \\ 0.0645 \end{bmatrix}$$

$$h_4 = \tanh \begin{bmatrix} 0.5578 \\ 0.0645 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5063 \\ 0.0644 \end{bmatrix}$$

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Output Calculation

$$y = W_{hy} h_4 + b_y$$

$$y = [1.5 \quad -1.2] \begin{bmatrix} 0.5063 \\ 0.0644 \end{bmatrix} + 0.5$$

$$y = (1.5 \times 0.5063) - (1.2 \times 0.06447) + 0.5$$

$$y = 0.7595 - 0.0773 + 0.5$$

$$= 1.1822$$

Apply sigmoid

$$\hat{y} = \sigma(y) = \frac{1}{1+e^{-1.1822}} = 0.765$$

66

$q = [2, 4]$

$K = [1, 0]$ $d_K = 4$

$V = [2]$

attention(q, K, v) =

$$\text{softmax}\left(\frac{q \cdot K^T}{\sqrt{dk}}\right)V$$

1. $(q \cdot K^T) = (2 \times 1) + (4 \times 0)$

$= 2$

2. $(\sqrt{dK})$

$\sqrt{dk} = \sqrt{4} = 2$

3. Scaled attention score

$$\text{Score} = \frac{q \cdot K^T}{\sqrt{dk}} = \frac{2}{2} = 1$$

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4 Apply softmax

$$\text{Softmax}([1]) = \frac{e^1}{e^1} = 1$$

5 Compute the final output

$$\text{Attention}(q, K, v) = 1 \cdot V$$

$$= 1 \cdot [2, 1]$$

$$= [2, 1]$$

$$\boxed{\text{Ans} = [2, 1]}$$

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Q9

Mathematical formulation

1. Input Representation

$$X = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix}$$

2. Linear Projection

$$Q = XW_Q \in R^{n \times d_K}$$

$$K = XW_K \in R^{n \times d_K}$$

$$V = XW_V \in R^{n \times d_V}$$

3. Scaled Attention Score

$$S = \frac{QK^T}{\sqrt{d_K}} \in R^{n \times n}$$

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For individual elements,

$$A_{ij} = \frac{\exp(s_{ij})}{\sum_{m=1}^{n}}$$

$$s_{ij} = \frac{q_i \cdot k_j^T}{\sqrt{d_k}}$$

4 Output Context Representation

$$\text{Attention}(q, k, v) = \text{Softmax}\left(\frac{qk^T}{\sqrt{d_k}}\right) v \in R^{n \times d_v}$$

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Q10 CNN

1. Convolution Layer

$$z(i,j) = (x * k)(i,j) + b$$
$$= \sum_m \sum_n x(i-m, j-n)\, k(m,n) + b$$

2. Activation Function

$$A(i,j) = ReLU(z(i,j))$$
$$= \max(0, z(i,j))$$

3. Pooling Layer (Max Pooling)

$$P(i,j) = \max_{(m,n) \in R} A(i+m, j+n)$$

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Q. Fully connected

$$y = Ww + b_{fc}$$

$$\hat{y} = \text{Softmax}(y) = \frac{e^{y_k}}{\sum_j e^{y_j}}$$

Q8

position : 0, 1

d = 9

Formula

$$PE_{(pos,2i)} = \sin\left(\frac{Pos}{10000^{2i/d}}\right)$$

$$PE_{(pos,2i+1)} = \cos\left(\frac{Pos}{1000^{2i/d}}\right)$$

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For i = 0 (dimension 0, 1):

$$w_0 = \frac{1}{10000^{0/4}} = \frac{1}{1} = 1$$

For i = 1 (dimension 2, 3):

$$w_1 = \frac{1}{10000^{2/4}} = \frac{1}{\sqrt{10000}} = \frac{1}{100}$$

$$= 0.01$$

Calculate

pos = 0

1. PE(0) (2i = 0):

$$PE(0,0) = \sin(0 \times 1) = \sin(0) = 0$$

2. PE(1) (2i + 1 = 1):

$$PE(0,1) = \cos(0 \times 1) = \cos(0) = 1$$

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3. Dimension 2 (2i=2):

$PE(0,2) = \sin(0 \times 0.01)$

$= \sin(0) = 0$

4. D 3 (2i+1=3);

$PE(0,3) = \cos(0 \times 0.01)$

$= \cos(0) = 1$

$PE(pos=0) = [0, 1, 0, 1]$

$Pos = 1$

1. Dimension 0 (2i=0):

$PE(1,0) = \sin(1 \times 1) = \sin(1 rad)$

≈ 0.8415

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2. Dimension 1 ($2i+1=1$)

$PE(1,1) = \cos(1 \times 1) = \cos(1 \text{ rad})$

≈ 0.5403

3. Dimension 2 ($2i+1=2$):

$PE(1,2) = \sin(1 \times 0.01) = \sin(0.01 \text{ rad}) \approx 0.0100$

4. Dimension 3 ($2i+1=3$):

$PE(1,3) = \cos(1 \times 0.01) = \cos(0.01 \text{ rad}) \approx 0.99995 \approx 1.000$

$PE(pos=1) \approx [0.8415, 0.5403, 0.0100, 1.0000]$

Ans:

$PE(pos=0) = [0, 1, 0, 1]$

$PE(pos=1) \approx [0.8415, 0.5403, 0.0100, 1.0000]$